

Case Report

Isolated scrotal skin laceration following dog bite complicated by post-traumatic chronic scrotal pain: A case report

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ABSTRACT

Background: Genital dog bite injuries are rare but potentially severe. While most literature-reported cases describe testicular injuries and significant tissue loss, isolated scrotal skin lacerations represent a less severe clinical entity.

Case presentation: We report a 32-year-old male with an isolated skin laceration of the scrotum due to a dog bite, without any testicular involvement on clinical and ultrasonographic evaluation. Debridement and primary closure with antimicrobial prophylaxis and tetanus-rabies immunization ensured favorable early outcomes. However, the patient returned at 6 months with significant chronic scrotal pain despite unremarkable repeat clinical and imaging examinations.

Conclusion: This case highlights that even in the absence of structural injury, dog bite trauma to the scrotum may lead to post-traumatic chronic scrotal content pain (CSCP) through neuropathic sensitization mechanisms. Early recognition and a multimodal pain management approach are essential for optimal long-term outcomes.

Introduction

Animal bites account for around 1% of all presentations in the emergency department. Dogs were responsible for 80–90% of the injuries. [1] The common anatomical sites involved are the face, head, neck, and extremities. Genital injuries are infrequent. Injuries to the scrotum from dog bites present a unique clinical challenge because of the bacterial flora of the dog oral cavity, the vascular nature of scrotal tissue, and the potential for serious complications such as infection, loss of tissue, and damage to the testicles.

Much of the literature reports significant tissue destruction, testicular avulsion, or penetrating testicular trauma due to scrotal dog bites. [2,3] Isolated scrotal skin laceration without deeper structural involvement is rarely reported; however, these represent an important clinical entity with the possibility of successful management with appropriate early intervention.

The management strategies for genital dog bites have evolved from the traditional approach of delayed wound closure into the contemporary approach of early primary repair, provided adequate debridement and infection prophylaxis are implemented. [4] This change represents an evolution in our understanding of wound healing dynamics and the effectiveness of antimicrobial therapy.

We report a case of an isolated scrotal skin laceration resulting from a dog bite in a 32-year-old man, with initial favorable surgical outcome but subsequent development of post-traumatic chronic scrotal pain at 6 months, emphasizing the importance of long-term follow-up even in apparently uncomplicated cases.

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Case presentation

Initial presentation and management

A 32-year-old male with no significant past medical history presented to our emergency department approximately 3 h post injury for a dog bite to the scrotum. The patient reported that he was assaulted by a stray dog in his neighborhood. The bite occurred with sufficient force to break through his clothes, with immediate bleeding from the genital area.

Preliminary evaluation found that the patient was hemodynamically stable, afebrile, and showed normal vital signs (BP 125/75 mmHg, pulse rate 82 bpm, and temperature 36.8 °C). Local examination showed a 4 cm laceration on the anterior part of the scrotum with irregular margins, indicative of a bite wound. Active bleeding was controlled with direct pressure. Both testes were present in their normal position, without tenderness, swelling, or abnormal texture. There was no hematoma or discoloration beyond the immediate laceration site. (Fig. 1).

A scrotal ultrasound confirmed bilateral normal testicular size, echogenicity, and vascularity. The tunica albuginea was intact bilaterally, with no rupture, hematoma, or hematocele. Doppler study demonstrated normal testicular blood flow. Superficial soft tissue edema was noted in the scrotal wall overlying the laceration. (Fig. 2).

Laboratory analysis showed a white blood cell count of 9200/ μ L, hemoglobin of 14.2 g/dL, and normal platelet count. The patient had an incomplete tetanus vaccination history, with the last dose administered more than 10 years prior.

The patient was taken to the operating room 6 h post-presentation. Under local anesthesia with sedation, the wound was irrigated with 2 L of normal saline solution and povidone-iodine antiseptic solution. Tissue cultures were obtained. Debridement of wound margins was performed, followed by approximation of the dartos muscle using 3-0 absorbable sutures, and skin closure in a sub-cuticular fashion using 4-0 absorbable sutures. (Fig. 3) No drain placement was required given the absence of deep tissue injury.

Antimicrobial prophylaxis consisted of IV ampicillin-sulbactam 3 g preoperatively, followed by oral amoxicillin-clavulanate 875/125 mg twice daily for 7 days. The tetanus toxoid (0.5 mL) was given intramuscularly. Considering that the biting dog is stray, post-exposure rabies prophylaxis was performed using human rabies immunoglobulin 20 IU/kg, where 50% was given locally while 50% intramuscularly in combination with the rabies vaccine on days 0, 3, 7, and 14. The patient was discharged after 24 h of admission under medications. The culture from the wound was positive for *S. aureus* and found to be sensitive to the antibiotic therapy. No signs of infection and wound dehiscence were seen during follow-up in one week. Full wound healing with minimal scarring and good appearance were noted during the one-month follow-up.

Late complication: Post-traumatic chronic scrotal pain

At six months after surgery, the patient complained of significant scrotal pain that had progressively worsened over the preceding



Fig. 1. Laceration of the anterior scrotal skin with irregular margins following dog bite.

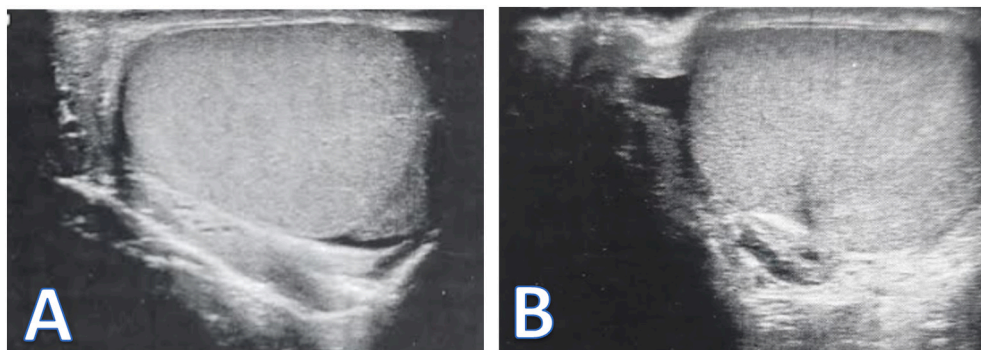


Fig. 2. Scrotal ultrasound demonstrating normal bilateral testes, intact tunica albuginea, and superficial soft tissue edema overlying the laceration.



Fig. 3. Post-debridement and primary closure of the scrotal laceration using absorbable sutures.

eight weeks. He complained of dull and aching pain with intermittent burning sensations, localized to the left hemiscrotum with occasional radiation to the ipsilateral groin and lower abdomen. His pain significantly affected his daily activities, physical exercises, and sexual activity. His pain had a VAS score of 7/10 at its worst.

Between the one-month postoperative review and the six-month presentation, the patient reported no intercurrent illness, trauma, urological symptoms, or new physical stressors. No sexual or occupational exposures to explain a *de novo* scrotal pathology were identified. The onset of pain at approximately four months post-injury, during a period of active scar maturation, is temporally consistent with the mechanism of perineural cicatricial fibrosis, which characteristically manifests weeks to months after apparent wound healing as collagen remodeling progressively encases adjacent nerve fibers.

Crucially, the patient denied any dysuria, fever, urethral discharge, or recent urinary tract infection. There was no history of trauma, heavy lifting, or new physical exertion. General and systemic examination was entirely unremarkable. Genital examination revealed a well-healed surgical scar over the anterior scrotum with no signs of local infection, abscess, or hematoma. Both testes were

palpable in their normal anatomical positions, with no appreciable tenderness on direct palpation, no epididymal thickening, and no palpable cord nodularity. The cremasteric reflex was intact bilaterally.

Repeat scrotal ultrasound with color-flow Doppler demonstrated bilaterally normal testes in terms of size, echotexture, and vascularity. No epididymo-orchitis, varicocele, hydrocele, spermatocele, or testicular mass was identified. The presence of surgical scar tissue in the scrotal wall was evident, but there were no signs of deep scar tissue or complications. The laboratory tests, including complete blood count and inflammatory markers such as C-reactive protein and erythrocyte sedimentation rate, were found to be within normal limits. Urinalysis and urine cultures were negative.

Based on the lack of an identifiable cause, the diagnosis of post-traumatic chronic scrotal content pain (CSCP) was confirmed, and the neuropathic component of the pain was related to peripheral nerve injury, which occurred during the initial trauma of the dog bite.

The patient was managed with a multi-modal approach, and the patient was put on low-dose amitriptyline (25 mg at night) and gabapentin (300 mg three times daily) for six weeks for neuropathic pain modulation. [9] NSAIDs were prescribed on an as-needed basis. Pelvic floor physiotherapy was recommended to address any myofascial component. Spermatic cord block with 0.25% bupivacaine was offered as both a diagnostic and therapeutic intervention.

At 3-month follow-up after initiating this pain management regimen (9 months post-surgery), the patient reported a significant reduction in pain intensity (VAS 3/10) with marked improvement in quality of life and return to normal daily activities. Long-term follow-up is being maintained to monitor for resolution or recurrence.

Discussion

This patient presentation reflects a successful early outcome followed by a clinically significant late complication—post-traumatic chronic scrotal content pain—arising 6 months after what initially appeared to be an uncomplicated isolated scrotal skin laceration from a dog bite. Several key points emerge from this case.

Epidemiology and clinical presentation

Dog bite injuries are responsible for a large number of animal bite injuries, with approximations of 740 per 100,000 population per year being reported in the literature. [1] Genital injuries from dog bites are uncommon, mostly reported in children and in those with restricted mobility. Adult injuries are usually severe, including testicular avulsion, penetration injuries, and gross soft tissue defects. [2,3] This particular injury is remarkable in that all deeper structures remained intact despite the bite mechanism.

The mechanism described in our patient, an unprovoked stray dog bite to the genitalia while clothed, although uncommon, is consistent with documented patterns of stray dog aggression. Several published cases of genital dog bite injuries in adults report similarly spontaneous or unprovoked attacks in outdoor public settings, lending biological plausibility to the reported mechanism.

Diagnostic evaluation

A comprehensive clinical evaluation represents a major priority but cannot always reliably rule out additional injuries. A misleading appearance of surface injuries can be present due to the intense crushing force of a dog's bite—ranging from 310 to 31,790 kPa depending on breed—which may reflect severe internal injury without significant disruption of the skin surface. [2] Scrotal ultrasound is therefore an essential diagnostic tool in all cases of scrotal trauma from dog bites, regardless of initial clinical findings. [5] Ultrasound reliably identifies testicular rupture, hematoma, hematocele, and vascular compromise, thereby guiding appropriate management decisions.

Management principles

The management of canine bite wounds has evolved from traditional delayed closure toward early primary repair. A 2025 systematic review and meta-analysis by Cui et al. demonstrated that primary closure is superior to delayed closure in reducing wound infection and improving cosmetic outcomes, particularly when presentation occurs within 10 h and prophylactic antibiotics are administered. [6] In our patient, all favorable criteria for primary closure were met: early presentation (6 h), absence of deep structural injury, limited devitalized tissue, thorough irrigation and debridement, and the inherent vascularity of genital skin which confers natural resistance to infection. Regarding antimicrobial prophylaxis, while the same meta-analysis found no statistically significant benefit of prophylactic antibiotics in dog bite wounds as a general rule, (NEW REF) expert consensus continues to recommend coverage for all genital dog bite injuries given the high local bacterial burden and proximity to perineal flora. Amoxicillin-clavulanate or ampicillin-sulbactam remain the agents of choice against the expected polymicrobial spectrum.

Infection prevention

This polymicrobial flora contributes to the large potential for infection. *Pasteurella multocida* is responsible for approximately half of dog bite infections, which manifest within 48 h [11]. Other pathogens include *Staphylococcus aureus*, *Staphylococcus intermedius*, *Capnocytophaga canimorsus*, anaerobes, and *Eikenella corrodens*. Prophylactic antibiotic therapy with either ampicillin-sulbactam or amoxicillin-clavulanate should be given for all genital dog bites to cover the expected polymicrobial spectrum [7]. The isolation of *Staphylococcus aureus* in the present patient, which was sensitive to the chosen antibiotic, confirmed the suitability of the antibiotic.

Immunization requirements

Tetanus and rabies prophylaxis are critical components of dog bite management. [1,8] Immunization against tetanus should be administered if five or more years have passed since the last booster, and passive immunization should be administered in cases where the immunization history is incomplete. Prophylaxis for rabies should be administered in cases where the animal involved in the dog bite cannot be observed for ten days, such as in stray dog bites. The World Health Organization recommends post-exposure prophylaxis, which includes infiltration of immunoglobulin at the site of injury, followed by administration of five doses of rabies vaccine. [1]

Post-traumatic chronic scrotal content pain: Pathophysiology and management

“Chronic scrotal pain syndrome” is defined as “scrotal discomfort lasting for more than 3 months, causing significant impairment in daily function.” CSP constitutes 2.5–4.8% of outpatient visits to a urologist. [9,10] Post-traumatic CSP from external genital bite injury is rare, and therefore, this case is an important addition to the literature.

- (1) **Post-Traumatic Peripheral Neuropathy.** Both the ilioinguinal nerve, which arises from the first lumbar nerve root, and the genitofemoral nerve, which arises from the first and second lumbar nerve roots, pass through the anterior scrotal wall in a superficial plane. Both nerves are therefore at risk for injury from the crush and tearing effects of a dog bite. [11] Axonal injury to the nerve fibers results in the symptoms of burning dysesthesia, spontaneous aching, and allodynia. The unilateral nature of chronic pain following a seemingly bilateral or anterior scrotal laceration is a recognized phenomenon in post-traumatic neuropathic conditions. The ilioinguinal and genitofemoral nerves display significant anatomical asymmetry in their terminal branching patterns, with variable density and depth of penetration into the scrotal wall between sides. A crushing bite force applied to the anterior scrotum may therefore produce asymmetric axonal injury depending on the precise trajectory of the wound margins relative to nerve course on each side. Furthermore, peripheral sensitization following nerve injury does not require symmetric injury; a lower threshold for central wind-up may be reached unilaterally when one side sustains greater axonal damage, even from a single wound crossing the midline. The left-sided predominance in our patient is consistent with the known anatomical variability of scrotal innervation and does not undermine the post-traumatic etiology.
- (2) **Perineural Cicatricial Fibrosis.** During scar maturation, contracting collagen progressively encases adjacent nerve fibers in a fibrotic matrix, generating sustained mechanical compression and ischemic nociception independent of the original injury. [11] This mechanism accounts for the characteristic delayed onset of pain weeks to months after apparent wound healing, consistent with our patient's timeline.
- (3) **Central Sensitization (“Central Wind-Up”).** Repetitive C-fiber and A-delta activation upregulates spinal NMDA receptors, establishing pain that is disproportionate to and independent of any ongoing peripheral stimulus. [9,11] This explains the high VAS score (7/10), normal imaging, and referred pain to the groin observed in our patient.

Diagnosis

Per the AUA 2025 Guideline, CSCP is a diagnosis of exclusion based on symptoms; imaging and laboratory studies serve only to rule out structural or infectious causes such as epididymo-orchitis, varicocele, or neoplasm. [8]

First-line therapy (conservative and pharmacological)

Non-pharmacological measures include scrotal support, warm sitz baths, activity modification, and pelvic floor physiotherapy. Neuropathic pharmacotherapy with amitriptyline (10–75 mg/night) and gabapentinoids (gabapentin 300–900 mg three times daily or pregabalin 75–150 mg twice daily) constitutes the cornerstone of first-line treatment. [10,12] In our patient, this combination reduced VAS from 7/10 to 3/10 over 3 months.

Second-line therapy (interventional)

Ultrasound-guided spermatic cord block (SCB) with 0.25–0.5% bupivacaine serves as both a diagnostic and therapeutic tool; a $\geq 50\%$ pain reduction confirms peripheral neuropathic etiology and predicts surgical success. [10] Trigger point injections, low-intensity extracorporeal shockwave therapy, and cognitive behavioral therapy may be added at this stage. [9,11]

Third-line therapy (surgical)

Microsurgical Denervation of the Spermatic Cord (MDSC) represents the gold standard for the treatment of refractory chronic scrotal content pain (CSCP), with a success rate ranging from 70 to 95%. This intervention should be considered for those patients showing a positive response to a spermatic cord block (SCB). [10] Targeted neurectomy of the ilioinguinal or genitofemoral nerve may be used to enhance the efficacy of MDSC when perineural fibrosis represents the major aetiologic cause. [11] Epididymectomy and orchiectomy should be considered as a last resort, reserving these interventions for those patients after exhaustive treatment and evaluation. [9]

The current case represents one of a limited number of reports detailing chronic scrotal content pain as a delayed complication of scrotal trauma. Patients should be counselled appropriately, with a low threshold for instituting neuropathic pain relief when symptoms develop.

Clinical implications

This case emphasizes two distinct clinical messages. First, not all scrotal dog bites result in catastrophic structural injury; when deeper structures are preserved, excellent short-term outcomes can be achieved with appropriate early management. Second, clinicians must remain vigilant for late functional complications, particularly chronic scrotal pain, even following apparently minor injuries with complete wound healing. Long-term follow-up protocols for scrotal dog bite injuries should be extended beyond the conventional wound-healing period to capture and manage delayed pain syndromes before they become refractory.

Establishing causation in post-traumatic chronic scrotal pain

Establishing cause and effect between a prior injury and subsequent chronic scrotal pain requires careful exclusion of alternative etiologies and demonstration of temporal, anatomical, and mechanistic plausibility. In our patient, all four criteria were satisfied: (i) temporal plausibility — pain emerged during the window of active scar maturation (4–6 months post-injury), consistent with perineural fibrosis; (ii) anatomical plausibility — the wound site overlies the terminal branches of the ilioinguinal and genitofemoral nerves; (iii) mechanistic plausibility — the crushing and tearing bite mechanism is sufficient to produce axonal injury; and (iv) exclusionary rigor — repeat clinical examination, scrotal ultrasound with Doppler, inflammatory markers, and urinalysis were all unremarkable, eliminating infectious, vascular, and structural causes. The absence of any intercurrent event between the one-month review and pain onset further strengthens the post-traumatic attribution.

Conclusion

Isolated scrotal skin lacerations from dog bites can be managed successfully with conservative surgery in the absence of testicular injury, yielding excellent short-term functional and cosmetic outcomes. However, as illustrated in this case, such injuries may be complicated by post-traumatic chronic scrotal content pain arising months after complete wound healing, in the absence of any identifiable structural, infectious, or vascular pathology. This late complication, likely mediated by neuropathic sensitization of the scrotal innervation, requires early recognition and a multimodal management strategy encompassing neuropathic analgesics, physiotherapy, and procedural interventions. Clinicians should maintain a high index of suspicion for occult injuries at initial presentation, and should establish long-term follow-up protocols that extend well beyond wound healing to identify and address delayed pain syndromes in this challenging clinical entity.

CRediT authorship contribution statement

Maachi Youssef: Writing – review & editing, Writing – original draft, Validation, Supervision, Investigation, Formal analysis, Data curation, Conceptualization. **Boustani Amine:** Methodology, Investigation. **Amine Salim Lalaoui:** Software, Resources, Project administration. **Slaoui Amine:** Visualization, Validation, Supervision. **Karmouni Tariq:** Visualization, Validation. **Koutani Abdelatif:** Visualization, Validation. **Elkhaider Khalid:** Visualization, Validation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

- [1] B. Karatas, E. Kapi, An unusual etiology of a scrotal dog bite injury and review of the literature, *Ulus. Travma Acil Cerrahi Derg.* 28 (4) (2022) 549–553, <https://doi.org/10.14744/tjtes.2020.66642>.
- [2] D. Saleh, D. Shaw, C.S. Biyani, A dog bite to the adult scrotum, *Can. Urol. Assoc. J.* 3 (5) (2009) E64–E66, <https://doi.org/10.5489/cuaj.1166>.
- [3] A. Boughaleb, R. Tariqi, H. El Bote, An unusual etiology of testicular avulsion: a case report, *Int. J. Surg. Case Rep.* 121 (2024) 109942, <https://doi.org/10.1016/j.ijscr.2024.109942>.
- [4] M. Javaid, L. Feldberg, M. Gipson, Primary repair of dog bites to the face: 40 cases, *J. R. Soc. Med.* 91 (8) (1998) 414–416, <https://doi.org/10.1177/014107689809100804>.
- [5] J.C. Buckley, J.W. McAninch, Use of ultrasonography for the diagnosis of testicular injuries in blunt scrotal trauma, *J. Urol.* 175 (1) (2006) 175–178, [https://doi.org/10.1016/S0022-5347\(05\)00048-0](https://doi.org/10.1016/S0022-5347(05)00048-0).
- [6] M. Cui, Y. Jia, Z. Chen, J. Qu, Z. Zhu, Y. Xu, S. Liu, R. Chen, Y. Shan, Primary closure and prophylactic antibiotics for treatment of traumatic wounds caused by mammals, a systematic review and meta-analysis, *World J. Emerg. Surg.* 20 (2025) 48, <https://doi.org/10.1186/s13017-025-00619-1>.
- [7] C.M. Gomes, L. Ribeiro-Filho, A.M. Giron, A.I. Mitre, E.R. Figueira, S. Arap, Genital trauma due to animal bites, *J. Urol.* 165 (1) (2001) 80–83, <https://doi.org/10.1097/00005392-200101000-00020>.
- [8] P.K. Stefanopoulos, A.D. Tarantzopoulou, Facial bite wounds: management update, *Int. J. Oral Maxillofac. Surg.* 34 (5) (2005) 464–472, <https://doi.org/10.1016/j.ijom.2005.04.001>.
- [9] H.H. Lai, M.A. Pontari, C.E. Argoff, et al., Diagnosis and Management of Male Chronic Pelvic Pain (chronic prostatitis/chronic pelvic pain syndrome and chronic scrotal content pain): AUA guideline, *J. Urol.* (2025), <https://doi.org/10.1097/JU.0000000000004566>.
- [10] S. Kravchick, S.J. Parekattil, G. Bratslavsky, et al., A simplified treatment algorithm for chronic scrotal content pain syndrome, *Curr Urol.* 18 (4) (2024) 251–256, <https://doi.org/10.1097/CU9.0000000000000240>.
- [11] O. Ergun, A. Gudeloglu, S.J. Parekattil, Management of chronic orchialgia: review of current clinical practice, *AME Med. J.* 8 (2023) 22, <https://doi.org/10.21037/amj-22-98>.
- [12] S.W. Leslie, H. Sajjad, L.E. Siref, Chronic testicular pain and Orchalgia, in: *StatPearls*, StatPearls Publishing, Treasure Island, FL, 2023.